

Technology

Eight stages between the utility and your glass.

This document is the engineering specification of the WPL purification stack. Every stage is independently certified, monitored, and serviced on an evidence-based schedule.

Code	Stage	Rating	Efficacy	Notes
01	Sediment pre-filter	5 µm	99%	Pleated polypropylene cartridge; pressure drop < 0.3 bar at 6 L/min.
02	Activated carbon block	0.5 µm	99%+	Coconut-shell carbon. Removes chlorine, chloramine, VOCs, taste, odor.
03	Reverse osmosis membrane	0.0001 µm	99.9%	Thin-film composite. NSF/ANSI 58 certified. PFAS, lead, fluoride, nitrate.
04	Deionization resin	Ionic	99.99%	Mixed-bed cation/anion. Output < 1 µS/cm. Specified for medical & lab.
05	Ultra-filtration ceramic	0.01 µm	99.99%	Hollow-fibre ceramic. Bacterial barrier post-tank.
06	UV-C sterilization	DNA	99.999%	265 nm dose. Always-on. Predictive lamp life monitoring.
07	Remineralization	—	—	Restores Ca and Mg ions for taste profile per installation.
08	Smart monitoring	—	Real-time	Inline TDS, flow, pressure. Streamed to private dashboard.

Operating envelope

Inlet pressure	1.0 – 6.0 bar
Inlet temperature	4 – 38 °C
Permeate flow (typical)	6 L/min @ 4 bar
Recovery ratio	50% (configurable 40–65%)
Power (UV + monitoring)	24 W continuous
Service interval	12 months scheduled / predictive
Warranty	10 years (parts, mechanical)

Certifications

NSF/ANSI 58 (RO performance) · NSF/ANSI 53 (lead, VOC) · NSF/ANSI 401 (emerging contaminants, PFOA/PFOS) · CE · WRAS Approved Material.

Specifying for your project

Each installation is configured against a fresh utility report. Contact the engineering team to begin a water test.

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